

WORKFLOW

Customer

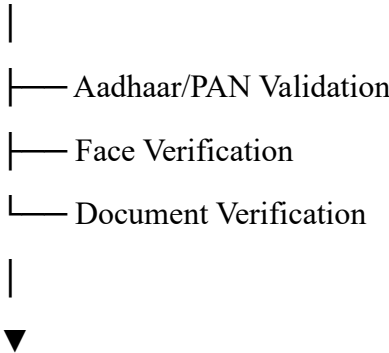


Loan Application

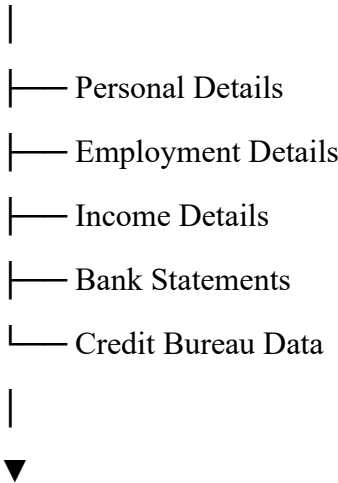
(Web / Mobile / Agent Portal)



Identity Verification (KYC)



Data Collection



Data Validation

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|— Missing Fields Check

|— Fraud Detection

|— Duplicate Customer Check

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Credit Assessment Engine

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|— Credit Score

|— Debt-to-Income Ratio

|— Risk Analysis

|— AI/ML Prediction

|— Eligibility Calculation

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Decision Engine

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|— Approved

|— Rejected

|— Manual Review

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Loan Offer Generation

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|— Loan Amount

|— Interest Rate

|— EMI

└─ Tenure

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Customer Acceptance

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Document Signing

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└─ eSign

└─ Agreement Generation

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Loan Disbursement

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└─ Bank Transfer

└─ Notification

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Loan Servicing

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└─ EMI Tracking

└─ Payment Reminders

└─ Penalty Calculation

└─ Customer Support

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Loan Closure

Module-wise workflow

Module	Responsibility
User Management	Registration, Login, Authentication
Loan Application	Customer submits loan request
KYC Module	Identity verification and document validation
Credit Assessment	Calculates eligibility and risk score
Decision Engine	Approves, rejects, or routes for manual review
Loan Management	Creates loan account and tracks status
Payment Module	EMI scheduling and payment processing
Notification Service	SMS, Email, Push notifications
Admin Dashboard	Monitor applications, approvals, reports
Analytics	Risk reports, portfolio analysis, fraud monitoring

Backend workflow

1. Customer submits the application.
2. Backend validates mandatory fields.
3. KYC service verifies identity.
4. Credit bureau APIs fetch credit history.
5. Risk engine calculates:
 - Credit score
 - Income eligibility
 - Existing liabilities
 - Fraud probability
6. Decision engine determines:

- Auto Approve
 - Auto Reject
 - Manual Review
7. Loan offer is generated.
 8. Customer accepts the offer.
 9. Agreement is generated and e-signed.
 10. Funds are disbursed to the customer's bank account.
 11. EMI schedule is created.
 12. Repayment tracking continues until loan closure.

Technologies commonly used

- **Frontend:** React, Angular, Flutter
- **Backend:** Java Spring Boot, .NET, Node.js
- **Database:** PostgreSQL, MySQL, Oracle, MongoDB
- **Authentication:** OAuth 2.0, JWT
- **Document Storage:** AWS S3, Azure Blob Storage
- **Messaging:** Kafka, RabbitMQ
- **AI/ML:** Python, TensorFlow, Scikit-learn
- **Reporting:** Power BI, Tableau

The workflow is typically:

1. Case creation
2. Customer onboarding
3. KYC verification
4. Document upload
5. Eligibility calculation
6. Credit bureau integration
7. Underwriter review (if required)
8. Approval workflow

9. Loan agreement generation

10. eSignature

11. Disbursement

12. Loan servicing

13. Loan closure